Project: Worked examples
Title: Insulation thickness (ISO 12241)
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1. Scope and method

The insulation thickness on a pipe (cylinder) or a flat surface that meets a heat-flow limit, a limit on the surface temperature of a hot surface and an anti-condensation requirement for a cold one, each solved separately and the governing one reported; and for a chosen thickness the heat flow, the surface temperature and PASS/FAIL per criterion. Steady state, one layer, no thickness table of any standard.

WHAT IT CALCULATES	RELATION
Pipe resistance per metre	$R = \ln(D_e/D_i)/(2\pi\cdot\lambda) + 1/(h_{se}\cdot\pi\cdot D_e)$, $D_e = D_i + 2t$; with the inner film and pipe wall $R_{oth} = 1/(h_i\cdot\pi\cdot D_p) + \ln(D_i/D_p)/(2\pi\cdot\lambda_w)$ added in series
Flat resistance per square metre	$R = t/\lambda + 1/h_{se}$ (+ $1/h_i + t_w/\lambda_w$)
Heat flow	$q = (\theta_i - \theta_a)/R$, positive from the process to the ambient
Surface temperature	$\theta_{se} = \theta_a + q/(h_{se}\cdot\pi\cdot D_e)$ (pipe), $\theta_a + q/h_{se}$ (flat)
Surface coefficient	h_{se} as given, or $h_{se} = h_c + h_r$ with $h_r = \varepsilon\cdot\sigma\cdot(T_s^2 + T_a^2)(T_s + T_a)$; h_c from free convection to air (laminar or turbulent, the larger), forced convection (cross-flow cylinder or flat plate) and their cube-law blend; solved with θ_{se}, which it depends on
Heat-loss limit	the thickness beyond which $\lvert q\rvert \le q_{max}$; surface-temperature limit (hot): $\theta_{se} \le \theta_{se,max}$; anti-condensation (cold): $\theta_{se} \ge \theta_{dew}$ + margin, θ_{dew} entered (for example from B2) or from the Magnus relation at θ_a and RH
Thickness solution	each criterion's thickness by scanning t from 0 to 1 m for the last broken point and bisecting the bracket after it; governing t = max over the selected criteria
Critical diameter	$D_{crit} = 2\lambda/h_{se}$; below it, insulation raises the heat flow until D_e passes D_{crit}

Basis: Thermal insulation for building equipment and industrial installations: calculation rules[1](formula route (clause numbers to be checked)); External forced, free and mixed convection correlations[2](7th ed., §7.2, §7.4, §9.6, §9.9 (to be checked)); Magnus saturation vapour pressure (shared with B7)[3](cl. 6 (clause number to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
geometry	Surface	pipe (cylinder)		assumption: A round pipe.	any
D_i	Pipe outer diameter, the inside of the insulation (pipe)	114	mm	drawing: M-PIP-021 rev B	10 … 1000 mm
θ_i	Process (fluid) temperature	120	°C	spec: SPEC-M-12 §3: heating flow temperature	-50 … 400 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
θ_a	Ambient air temperature (also the radiation sink)	15	°C	spec: SPEC-M-12 §2: ambient design temperature	-20 … 50 °C
λ	Thermal conductivity of the insulation	0.04	W/(m·K)	assumption: Mineral wool; take the maker's value at the mean temperature (W3 material)	0.015 … 0.2 W/(m·K)
h_{mode}	Where the outer surface coefficient comes from	convection + radiation (correlation)		assumption: Surface coefficient computed from the correlation.	any
h_{se}	Outer surface coefficient h_{se} (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)	10	W/(m²·K)	assumption: Not used: h_se is computed.	3 … 40 W/(m²·K)
eps	Emissivity of the outer surface (for the correlation; with h_{se} as given it only judges that value against condensation on a cold surface)	0.1		assumption: Bright aluminium cladding; replace with the cladding maker's emissivity.	0.05 … 1
v_{wind}	Air speed over the surface (correlation; 0 = still air)	0	m/s	assumption: Still air, indoors.	0 … 20 m/s
L_{surf}	Height or flow length of the flat surface (correlation, flat)	1	m	assumption: Not used: a pipe.	0.1 … 10 m
$q_{max,m}$	Heat-loss limit per metre of pipe (generic default of 25 W/m, replace with your project limit; blank = not a criterion)	60	W/m	spec: SPEC-M-12 §5: heat loss not to exceed 60 W/m	1 … 1000 W/m
$\theta_{se,max}$	Highest surface temperature, for a hot surface (blank = not a criterion)	60	°C	spec: SPEC-M-12 §6: personnel protection, surface not above 60 °C	30 … 70 °C
RH_a	Relative humidity of the ambient air (anti-condensation)	60	%	assumption: Not used: a hot pipe has no condensation criterion.	20 … 95 %
t_{chosen}	Chosen insulation thickness (blank = report at the required thickness, no checks; with it, nothing blocks)	50	mm	spec: Pipework schedule M-SCH-021: 50 mm sections	0 … 300 mm

3. Assumptions

A1. Steady one-dimensional heat flow through a single homogeneous layer of constant conductivity; no thermal bridges at supports, no joints, no moisture in the insulation. λ is taken at the value entered: use the value at the mean insulation temperature shown on the sheet (optionally noted in 'Temperature the conductivity applies at'). (Formula route of the standard.)
A2. With the correlation, $h_{se} = h_c + h_r$. Radiation: $h_r = \varepsilon \cdot \sigma \cdot (T_s^2 + T_a^2)(T_s + T_a)$ to surroundings at the ambient air temperature. Free convection to air: Holman's simplified relations at atmospheric pressure, the larger of the laminar and turbulent forms (horizontal cylinder 1.32(ΔT/D)^0.25 and 1.24ΔT^(1/3); vertical surface 1.42(ΔT/L)^0.25 and 1.31ΔT^(1/3)). Forced convection: Churchill–Bernstein for a cylinder in cross-flow, the laminar/mixed flat-plate relations along a plate; air properties from fits at the film temperature, Pr = 0.71; blended with free convection as $(h_{nat}^3 + h_{forced}^3)^{(1/3)}$. Indicative, not the standard's own correlation. (Incropera 7th ed. §7.2, §7.4, §9.6 free convection and §9.9 mixed convection (section numbers to be checked against the owner's copy); the free-convection constants are Holman's, Heat Transfer; the standard's own coefficients are not reproduced.)
A3. The dew point is entered (for example from B2) or is the Magnus relation at θ_a and RH, the same closed form B7 uses. Below 0 °C the relation is over ice, so RH is then taken relative to ice, whereas meteorological RH is usually relative to water: enter the dew point for sub-zero air. A margin above it is the user's. (Derived; B7 shares the function.)
A4. Pipes are treated as horizontal for free convection and flat surfaces as vertical; air properties are fits (conductivity within about 3 %, viscosity within about 4 % over 250–400 K), Pr = 0.71. (Stated.)
A5. The required thickness is the least thickness beyond which the criterion holds for every thicker layer up to 1 m, which is what matters when thin insulation on a small pipe raises the heat flow (critical diameter). (Stated.)
A6. No thickness table is used or shipped: not BS 5422, not the tables of any standard. The conductivity is the user's input, from a W3 material or the maker's data. (No standard's table is reproduced.)
A7. geometry (Surface) is taken as pipe (cylinder). (A round pipe.)
A8. λ (Thermal conductivity of the insulation) is taken as 0.04 W/(m·K). (Mineral wool; take the maker's value at the mean temperature (W3 material))
A9. h_{mode} (Where the outer surface coefficient comes from) is taken as convection + radiation (correlation). (Surface coefficient computed from the correlation.)
A10. h_{se} (Outer surface coefficient h_{se} (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)) is taken as 10 W/(m²·K). (Not used: h_{se} is computed.)
A11. eps (Emissivity of the outer surface (for the correlation; with h_{se} as given it only judges that value against condensation on a cold surface)) is taken as 0.1. (Bright aluminium cladding; replace with the cladding maker's emissivity.)
A12. v_{wind} (Air speed over the surface (correlation; 0 = still air)) is taken as 0 m/s. (Still air, indoors.)
A13. L_{surf} (Height or flow length of the flat surface (correlation, flat)) is taken as 1 m. (Not used: a pipe.)
A14. RH_a (Relative humidity of the ambient air (anti-condensation)) is taken as 60 %. (Not used: a hot pipe has no condensation criterion.)

4. Working

[11] t_q = t such that |q| ≤ 60 W/m for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 23.374 mm
Required thickness, heat-loss limit

[12] t_{se} = t such that θ_{se} ≤ 60 °C for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 7.8423 mm
Required thickness, surface-temperature limit

[13] thickness = max over the selected criteria [1]
= t_q = 23.374 mm, t_{se} = 7.8423 mm
= 23.374 mm
Governing required thickness (heat-loss limit)
[15] $D_e = D_i + 2t$
= 114 mm + 2 × 50 mm
= 214 mm
Outer diameter of the insulation, chosen thickness
[16] $R_{ins} = \ln(D_e/D_i)/(2\pi\cdot\lambda)$ [1]
= $\ln(214/114)/(2\pi \times 0.04)$
= 2.5058 m·K/W
Resistance of the insulation
[17] h_c = free convection to air, blended with forced convection [2](7th ed. §7, §9.6, §9.9)
= horizontal cylinder, Holman; Churchill–Bernstein if the wind blows
= 3.6748 W/(m²·K)
Convective surface coefficient at the surface temperature
[18] $h_r = \varepsilon\cdot\sigma\cdot(T_s^2 + T_a^2)(T_s + T_a)$ [2](7th ed. §1.2)
= ε = 0.1, θ_{se} = 27.855 °C
= 0.58007 W/(m²·K)
Radiative surface coefficient
[19] $h_{se} = h_c + h_r$ [1]
= 3.6748 + 0.58007
= 4.2549 W/(m²·K)
Outer surface coefficient used
[20] $R_{se} = 1/(h_{se}\cdot\pi\cdot D_e)$ [1]
= $1/(4.2549 \times \pi \times 0.214$ m)
= 0.34958 m·K/W
Resistance of the outer surface
[21] $R = R_{oth} + R_{ins} + R_{se}$ [1]
= 0 + 2.5058 + 0.34958
= 2.8554 m·K/W
Total resistance per metre
[22] $q_{per,m} = |\theta_i - \theta_a| / R$ [1]
= 105 K / 2.8554
= 36.773 W/m
Heat flow per metre of pipe (heat loss, magnitude)
[23] $\theta_{se} = \theta_a + q/(h_{se}\cdot\pi\cdot D_e)$ [1]
= 15 °C + 36.773 / 2.8606
= 27.855 degC
Outer surface temperature
[24] $\theta_m = (\theta_i + \theta_{se})/2$
= (120 + 27.855)/2
= 73.927 degC
Mean temperature of the insulation: take λ at this temperature
[25] $D_{crit} = 2\lambda/h_{se}$ at the bare surface [1]
= 2 × 0.04 / 8.1897
= 9.7683 mm
Critical diameter at the bare-surface h_{se}: below it, insulation raises the heat flow
[26] $u_q = |q| / q_{max}$
= $|q| \le 60$ W/m
= 0.61288
Utilisation, heat-loss limit

[27] $u_{se} = (\theta_{se} - \theta_a)/(\theta_{se,max} - \theta_a)$
$= \theta_{se} \le 60\ °C$
$= 0.28567$
Utilisation, surface-temperature limit

5. Results

SYMBOL	VALUE	DESCRIPTION
thickness	23.37 mm	Governing required thickness
$q_{per,m}$	36.77 W/m	Heat flow per metre (heat loss)
θ_{se}	27.85 °C	Outer surface temperature
t_q	23.37 mm	Required thickness, heat-loss limit
t_{se}	7.842 mm	Required thickness, surface-temperature limit
θ_m	73.93 °C	Mean insulation temperature
R	2.855 m·K/W	Total resistance
D_{crit}	9.768 mm	Critical diameter (at the bare-surface h_{se})
h_c	3.675 W/(m²·K)	Convective surface coefficient
h_r	0.5801 W/(m²·K)	Radiative surface coefficient
h_{se}	4.255 W/(m²·K)	Outer surface coefficient
governing	"heat-loss limit"	Governing criterion
u_q	0.61	Utilisation, heat-loss limit (governing)
u_{se}	0.29	Utilisation, surface-temperature limit

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Chosen thickness against the heat-loss limit	$u_q = 0.61$	$u_q \le 1$	0.61	[1]
PASS	Chosen thickness against the surface-temperature limit	$u_{se} = 0.29$	$u_{se} \le 1$	0.29	[1]

7. Applicability

WARNING records: tabulated values not taken from a reference record are λ (Material properties); h_{se} (Outer surface coefficient); eps (Surface emissivity); $q_{max,m}$, $\theta_{se,max}$ (Insulation criteria). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

Template bs.insulation@1.1.3 Content hash 1838108ae7379e74 Engine calc-engine/1.0.0
Known-answer cases passed 14/14 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. ISO 12241.
2. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
3. ISO 13788.